

# Database Management Systems



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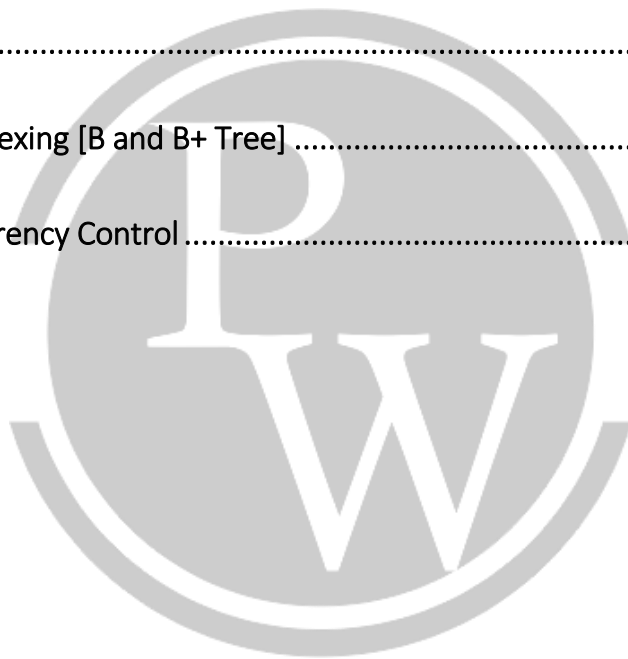
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# Database Management Systems

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# 1

# DATABASE DESIGN

## 1.1 Introduction

A database is an organized collection of structured data or related data.

## 1.2 Limitations of File System

- The physical details of the data to access the database are managed by the user.
- File system can be used to manage small database.
- When database is large, the operating system fails to control the concurrency.
- Only single user can access the whole data of the file system at a time.

## 1.3 Integrity Constraints Of RDBMS

- According to codd, data in database file must be stored in tabular format (set of row's and column's).
- According to codd, no two row's/records of the table should be equal.

### 1.3.1 Arity

Number of attributes of database table.

### 1.3.2 Cardinality

Number of records of database table.

## 1.4 Keys in database

- **Candidate Key:** A minimal set of attributes that differentiate records/row's of DB table uniquely.
- **Primary Key :** One of the candidate key whose field value cannot be null.
- **Simple Candidate Key:** A candidate key with only one attribute.
- **Compound Candidate Key:** A candidate key with atleast two attributes.
- **Overlapped Candidate Key:** Two or more candidate key with some common attribute.
- **Prime attribute:** The attribute that belongs to some candidate keys of a relation.
- **Non-prime attribute:** The attribute that does not belongs to any of the candidate keys of the relation.

**Example :**

Consider a relation R (ABCDE)

1. Assume candidate key : {AB, BC}



2. The above candidate keys are compound and overlapped.
3. Prime attribute {A, B, C}
4. Non-prime attribute : {D, E}

### 1.4.1 Difference between Primary key and Alternative key

| Primary Key                                                                                                                                                          | Alternative Key                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ol style="list-style-type: none"> <li>Any one candidate key whose field value can not be null.</li> <li>Atmost one primary key allowed for any DB table.</li> </ol> | <ol style="list-style-type: none"> <li>All candidate key of relation except primary key, whose field value can be null.</li> <li>Many (o or more) alternative keys are allowed for DB table.</li> </ol> |

**Note:**

For any RDBMS table there must be atleast one candidate key, whose field value can not be null.

(Unique + Not Null)  $\neq$  Primary key

### 1.4.2 Super key

The set of attributes which can differentiate records/tuples uniquely or super set of candidate key.

**Example 1:** R (ABCD) with CK = {A}

$$\therefore \text{Super key} = \text{CK} \cdot [\text{Any subset of other attributes (BCD)}]$$

$$= A \cdot [2^3] = 8 \text{ Super key.}$$

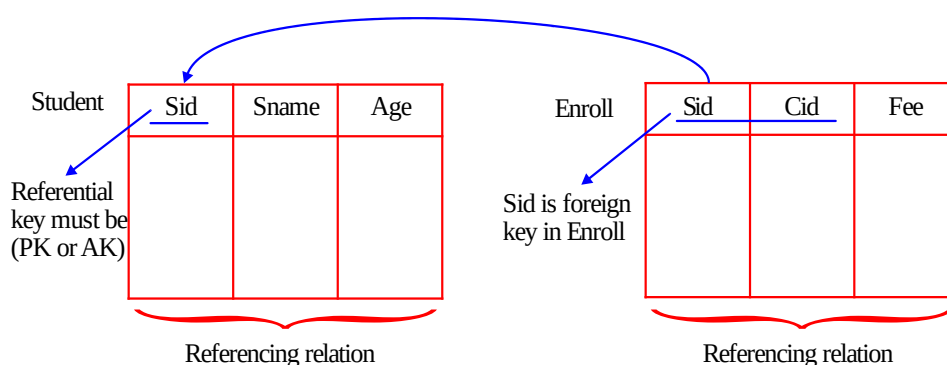
**Example 2 :** Let R be the relational schema with n-attibutes, R (A<sub>1</sub>, A<sub>2</sub>, ..... A<sub>n</sub>) then number of superkeys possible:

- (I) With (A<sub>1</sub>) as candidate key :  $2^{n-1}$
- (II) With {A<sub>1</sub>, A<sub>2</sub>} as candidate key :  $2^{n-1} - 2^{n-2} + 2^{n-1}$
- (III) With {A<sub>1</sub>A<sub>2</sub>, A<sub>3</sub>A<sub>4</sub>} as candidate key:  $2^{n-2} - 2^{n-4} + 2^{n-2}$
- (IV) The maximum number of super keys possible when each attribute of R is candidate key :  $2^{n-1}$
- (V) The minimum number of super key possible when all the attributes combined form single candidate key: 1 {A<sub>1</sub>A<sub>2</sub> ..... A<sub>n</sub> : candidate key}

## 1.5 Referential Integrity Constraints

### 1.5.1 Foreign key

Foreign key is a set of attributes that references primary key or alternative key of the same relation or other relation.



### Referenced Relation



1. **Insertion** : No violation
2. **Deletion** : [May cause violation]
  - (a) On delete no action : Means if it cause problem on delete then deletion is not allowed on table.
  - (b) On delete cascade : If we want to delete primary key value from referenced table then it will delete that value from referencing table also.
  - (c) On delete set null : If we want to delete primary key value from referenced table then it will try to set the null values in place of that value in referencing table.

**Note:**

If foreign key field is not null attribute then “On delete set null” is same as” on delete no action.”

3. **Updation** : [May cause violation]
  - (a) On update no action
  - (b) On update cascade
  - (c) On update set null

**Referencing Relation**

1. **Insertion** : [May cause violation]
2. **Deletion** : No violation
3. **Updation** : [May cause violation]

**Note**

If integrity violation occurs because of insertion or updation in referencing table then restrict insertion and updation.

**Example:**

| P.K |   | F.K                                                                      |  |
|-----|---|--------------------------------------------------------------------------|--|
| A   | B |                                                                          |  |
| 2   | 4 | B is foreign key referencing<br>A · delete (2, 4) and on delete cascade. |  |
| 3   | 4 |                                                                          |  |
| 4   | 5 |                                                                          |  |
| 5   | 4 |                                                                          |  |
| 6   | 2 |                                                                          |  |

So, If we delete (2, 4) then PK “2”. gets deleted from the table and all the tuples in which B is referencing PK.2” also gets deleted.

| A | B |
|---|---|
| 3 | 4 |
| 4 | 5 |
| 5 | 4 |

} Result

## 1.6 Database Design Goals

1. 0% redundancy
2. Loss-less join
3. Dependency preservation